Lecture 23 (Duality Introduction)

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Instead of solving the original LP directly, it is sometimes helpful to solve a related LP called the dual LP. This will be illustrated below:

Duality

Consider the L.P. below:

$$\text{minimize } 3x_1 + 4x_2$$

s.t.

$$2x_1 + 3x_2 \geq 13$$

$$1x_1 + 0x_2 \geq 2$$

$$0x_1 + 1x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

Then we can bound the objective function by taking a linear combination of the constraints. For example, suppose we take the following:

$$3\,(x_1 \geq 2)$$

$$3x_1 \geq 6$$

and

$$4\,(x_2 \geq 5)$$

$$4x_2 \geq 20$$

so if we add both of these scaled constraints, we see

$$3x_1 + 4x_2 \geq 20 + 6 = 26$$

Alternatively, we can add all three constraints and see that:

$$(2x_1 + 3x_2) + (x_1) + (x_2) \geq 13 + 2 + 5 = 20$$

Either way, we get lower bounds on the objective. Any feasible x_1, x_2 must yield an objective function that is greater than or equal to 20 and 26. If we try to make this left hand sum as large as possible, we get a related LP given below:

$$\text{maximize } 13y_1 + 2y_2 + 5y_3$$

s.t.

$$2y_1 + y_2 \leq 3$$

$$3y_1 + y_3 \leq 4$$

$$y_1, y_2, y_3 \geq 0$$

Here, each y_i corresponds to the ith constraint of the original LP. Each constraint in the new LP corresponds to a variable in the original LP. The objective in the new LP comes the the right hand side of the constraints in the original LP and the right hand side of the constraints in the new LP come from the coefficients in the objective of the original LP. We call this new LP the **Dual** of the original LP. Rather than using the term, "original LP", most authors use the term, "**Primal**" LP, so we will call the original LP the primal from here on out. Let's give a few more examples of this conversion process before stating the full rule set.

Ex 1

Consider the LP

$$\text{maximize } x_1 + 2x_2 + 3x_3$$

s.t.

$$1x_1 + 2x_2 + 0x_3 \leq 4$$

$$1x_1 + 0x_2 + 1x_3 \leq 3$$

$$0x_1 + 1x_2 + 4x_3 \leq 1$$

$$x_1, x_2, x_3 \geq 0$$

This time, the maximize function turns into a minimize function. The constraints are combined in linear combinations to give an upper bound on the maximize function. We will let each y_i correspond to the ith constraint of the primal and we will let each x_j correspond to the jth constraint in the dual.

$$\text{minimize } 4x_1 + 3x_2 + x_3$$

s.t.

$$1y_1 + 1y_2 + 0y_3 \geq 1$$

$$2y_1 + 0y_2 + 1y_3 \geq 2$$

$$0y_1 + 1y_2 + 4y_3 \geq 3$$

$$y_1, y_2, y_3$$

Ex 2

What if we are in standard vector form?

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

Yields the dual

$$\text{maximize } \vec{y} \cdot \vec{b}$$

s.t.

$$\vec{y}^T A \leq \vec{c}^T$$

$$\vec{y} \in \mathbb{R}^n$$

Note here we are taking the transposes for the dimensions to work out. If we take the transpose of both sides of the constraint, we see

$$\left(\vec{y}^T A \leq \vec{c}^T\right)^T$$

$$\left(\vec{y}^T A\right)^T \leq \vec{c}^{T^T}$$

$$A^T \vec{y} \leq \vec{c}$$

So you will often see the dual written as

$$\text{maximize } \vec{y} \cdot \vec{b}$$

s.t.

$$A^T \vec{y} \leq \vec{c}$$

$$\vec{y} \in \mathbb{R}^n$$

Further, as we have an equality constraint in the original problem, we will get $\vec{y} \in \mathbb{R}^n$. This is because we can multiply any equality constraint by any real number and still get an equality constraint. We do not have to worry about inequalities reversing, so we can take linear combinations without issue.

Ex 3

Using a similar idea, we see the primal

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} \geq \vec{b}$$

$$\vec{x} \geq 0$$

yields the dual

$$\text{maximize } \vec{y} \cdot \vec{b}$$

s.t.

$$A^T \vec{y} \leq \vec{c}$$

$$\vec{y} \geq 0$$

Notice in all of the example so far, we have had the inequalities in the constraint point "the right way". We were always able to add the constraints without issue, and we are approaching the objective function from the correct direction when we are attempting to bound it. What if the inequalities point "the wrong way"?

Ex 4

Consider the LP below:

$$\text{minimize } x_1 + 2x_2$$

s.t.

$$x_1 + x_2 \leq 3$$

$$0x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Notice that if were to add the inequalities directly, we would get

$$x_1 + 2x_2 \leq 7$$

This doesn't tell us much about the minimum. If we had negative coefficients however, we can learn quite a bit about the original LP. As such, we will modify require the appropriate coefficients be negative. This yields the dual below:

$$\text{maximize } 3y_1 + 4y_2$$

s.t.

$$y_1 + 0y_2 \leq 1$$

$$y_1 + y_2 \leq 4$$

$$y_1, y_2 \leq 0$$

Ex 5

Suppose we have a combination of constraints:

$$\text{minimize } x_1 + 2x_2 + 3x_3$$

s.t.

$$-x_1 + 3x_2 + 0x_3 = 5$$

$$2x_1 - x_2 + 3x_3 \geq 6$$

$$0x_1 + 0x_2 + 1x_3 \leq 4$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \in \mathbb{R}$$

yields the dual

$$\text{maximize } 5y_1 + 6y_2 + 4y_3$$

s.t.

$$-y_1 + 2y_2 + 0y_3 \leq 1$$

$$3y_1 - y_2 + 0y_3 \geq 2$$

$$0y_1 + 3y_2 + y_3 = 3$$

$$y_1 \in \mathbb{R}, y_2 \geq 0, y_3 \leq 0$$

Here, whether x_i is ≥ 0, ≤ 0, or $\in \mathbb{R}$ will dictate what type of constraint the appears in the dual. Similarly, where the primal constraints is $\geq$,$\leq$, or $=$ will determine what the corresponding restriction on y_i is.

Summary

Here is a summary of the overall conversion rules:

Primal	Transforms to	Dual
minimize	$\rightarrow$	maximize
$A\vec{x} \geq \vec{b}$	$\rightarrow$	$\vec{y} \geq 0$
$A\vec{x} \leq \vec{b}$	$\rightarrow$	$\vec{y} \leq 0$
$A\vec{x} = \vec{b}$	$\rightarrow$	$\vec{y} \in \mathbb{R}^n$
$\vec{x} \geq 0$	$\rightarrow$	$A^T\vec{y} \leq \vec{c}$
$\vec{x} \leq 0$	$\rightarrow$	$A^T\vec{y} \geq \vec{c}$
$\vec{x} \in \mathbb{R}^n$	$\rightarrow$	$A\vec{y} = \vec{c}$

Theorem

If we transform the dual LP into the equivalent minimization problem, find the dual of this new problem, then transform this back into the minimization problem, then we get back the primal LP. (This is a lot to write down and is a mouthful. The result is summarized in the diagram below).

$$\begin{array}{ccc} \text{Primal Minimization LP} & \xleftarrow{\text{Equivalent Minimization LP}} & \text{Primal Maximization LP} \\ \text{Take the Dual} \downarrow & & \uparrow \text{Take the Dual} \\ \text{Dual Maximization LP} & \xrightarrow{\text{Equivalent Minimization LP}} & \text{Dual Minimization LP} \end{array}$$

This theorem is often summarized as "The dual of the dual is the primal".

Proof

We will show this for 3 constraints and variables (one of each type). If you have more constraints and variables, a similar proof is used, but with the appropriate bock matrix rules. Consider the primal LP below:

$$\text{minimize } c_1x_1 + c_2x_2 + c_3x_3$$

s.t.

$$A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \geq b_1$$

$$A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \leq b_2$$

$$A_{31}x_1 + A_{32}x_2 + A_{33}x_3 = b_3$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \in \mathbb{R}$$

This turns into the dual maximization problem:

$$\text{maximize } b_1y_1 + b_2y_2 + b_3y_3$$

s.t.

$$A_{11}y_1 + A_{21}y_2 + A_{31}y_3 \leq c_1$$

$$A_{12}y_1 + A_{22}y_2 + A_{32}y_3 \geq c_2$$

$$A_{13}y_1 + A_{23}y_2 + A_{33}y_3 = c_3$$

$$y_1 \geq 0, y_2 \leq 0, y_3 \in \mathbb{R}$$

To convert to the equivalent minimization problem, we first replace each y_i with $-y_i'$. This yields

$$\text{maximize } -b_1y_1' - b_2y_2' - b_3y_3'$$

s.t.

$$-A_{11}y_1' - A_{21}y_2' - A_{31}y_3' \leq c_1$$

$$-A_{12}y_1' - A_{22}y_2' - A_{32}y_3' \geq c_2$$

$$-A_{13}y_1' - A_{23}y_2' - A_{33}y_3' = c_3$$

$$-y_1' \geq 0, -y_2' \leq 0, -y_3' \in \mathbb{R}$$

Now, if we multiply each constraint and the objective by -1, we convert this into the equivalent minimization problem:

$$\text{minimize } b_1y_1' + b_2y_2' + b_3y_3'$$

s.t.

$$A_{11}y_1' + A_{21}y_2' + A_{31}y_3' \geq -c_1$$

$$A_{12}y_1' + A_{22}y_2' + A_{32}y_3' \leq -c_2$$

$$A_{13}y_1' + A_{23}y_2' + A_{33}y_3' = -c_3$$

$$y_1' \leq 0, y_2' \geq 0, y_3' \in \mathbb{R}$$

Now, we take the dual of this to yield

$$\text{maximize } -c_1x_1 - c_2x_2 - c_3x_3$$

s.t.

$$A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \geq b_1$$

$$A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \leq b_2$$

$$A_{31}x_1 + A_{32}x_2 + A_{33}x_3 = b_3$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \in \mathbb{R}$$

Finally, we convert the objective into the equivalent statement to recover the primal LP.

$$\text{minimize } c_1x_1 + c_2x_2 + c_3x_3$$

s.t.

$$A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \geq b_1$$

$$A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \leq b_2$$

$$A_{31}x_1 + A_{32}x_2 + A_{33}x_3 = b_3$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \in \mathbb{R}$$